Year 7
Mathematics

Unit 1: Analysing and Displaying Data

1.1 Averages and Range

Averages and Range

STARTER · 5 MIN

Date: _____ **Title: Averages and Range**

LEARNING OBJECTIVES

- To calculate and interpret the mean, median, mode and range of a set of data

STARTER *Answer in your book. No calculators.*

1. Put in order: 34, 19, 42, 27
2. Work out 18 + 24 + 21
3. Work out 30 ÷ 5
4. Work out 27 − 16

Starter — Answers

STARTER

Tick or correct in green pen

1. **19, 27, 34, 42**

2. **63**

3. **6**

4. **11**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify the mode, median and range from a set of data	☐
SILVER	I can calculate the mean accurately	☐
GOLD	I can compare and interpret sets of data using averages and range	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Averages and Range

INTRODUCTION · 3 MIN

An average represents a set of data. The range shows how spread out it is.

Mode

The value that appears most often.

4, 7, 4, 5 → mode = 4

Median

The middle value when the data is in order.

2, 5, 9 → median = 5

Mean

Total of all the values ÷ number of values

6, 8, 4 → 18 ÷ 3 = 6

Range

Highest value − lowest value

16, 27, 20 → 27 − 16 = 11

Examples 1 and 2 — Mode and Range

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Mode = most common value Range = highest − lowest

Example 1

A teacher recorded the number of books read by 10 students:

4, 7, 5, 4, 8, 6, 4, 7, 5, 4

Work out the mode.

Example 2

The temperatures, in °C, recorded over 7 days were:

18, 24, 21, 16, 27, 20, 23

Work out the range.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Mode: the most common value

The number 4 appears most often.

Mode = 4 books

Tip: there can be more than one mode.

Example 2

Range: highest − lowest

Highest value = 27

Lowest value = 16

Range = 27 − 16

= 11 °C

Tip: the range is one number, not 16–27.

Task 1

PRACTICE · 9 MIN

BRONZE *Find the mode or the range*

Hint Mode = most common value Range = highest − lowest

1. Find the mode:
   6, 8, 5, 6, 9, 6, 7

2. Find the range:
   34, 19, 42, 27, 51, 23, 38

3. Find the mode:
   1, 1, 5, 4, 0, 5, 5

4. Find the range:
   15, 8, 22, 11, 19

5. Find the mode:
   12, 15, 14, 15, 13, 15, 12

★ Find the mode and the range:
   7, 3, 9, 7, 4, 7, 10

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 **Mode = 6**

2 **Range = 51 − 19**
= 32

3 **Mode = 5**

4 **Range = 22 − 8**
= 14

5 **Mode = 15**

★ **Mode = 7**
Range = 10 − 3
= 7

Examples 3 and 4 — The Median

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Put the values in order first Median = middle value

Example 3

Sheena recorded the rainfall in mm each day for a week.
Here are her results:
72, 15, 9, 80, 51, 38, 22

Work out the median daily rainfall.

Example 4

Paul recorded the number of goals scored by his football team last month:
3, 2, 0, 1, 1, 3

Work out the median number of goals scored.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Write the values in order

9, 15, 22, 38, 51, 72, 80

The middle (4th) value is 38

Median = 38 mm

Tip: 7 values, so the 4th is the middle.

Example 4

Write the values in order

0, 1, 1, 2, 3, 3

Two middle values: 1 and 2

Median = (1 + 2) ÷ 2

= 1.5 goals

Tip: even number of values: halfway.

Task 2

PRACTICE · 9 MIN

SILVER

Work out the median of each set of data

Hint Put the values in order first
Median = middle value

1. Books read by 6 students:
   34, 21, 48, 27, 39, 16

2. Goals scored in 7 matches:
   28, 45, 17, 36, 52, 21, 40

3. Find the median:
   5, 9, 2, 7, 4

4. Find the median:
   13, 8, 10, 4, 11, 6

5. Find the median:
   1, 1, 5, 4, 0, 5, 5

★ Find the median:
   25, 31, 18, 40, 22, 35, 29, 19

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 **Median = (27 + 34) ÷ 2**
= 30.5 books

2 **Median = 36 goals**

3 **Median = 5**

4 **Median = (8 + 10) ÷ 2**
= 9

5 **Median = 4**

★ **Median = (25 + 29) ÷ 2**
= 27

Examples 5 and 6 — Mean and Comparing Data

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint
Mean = total ÷ number of values
Smaller range = more consistent

Example 5

(2 marks)

The number of books read by 5 students is:
6, 8, 4, 7, 5

Work out the mean number of books.

Example 6

(2 marks)

Points scored in 5 games:
Ali: 6, 9, 7, 8, 7
Omar: 2, 12, 5, 10, 4

Use the median and range to compare Ali and Omar.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Mean = total ÷ number of values

Step 1: add all the values

Total = 6 + 8 + 4 + 7 + 5

= 30

Step 2: divide by 5

Mean = 30 ÷ 5

= 6 books

Tip: adding a 0 makes the mean smaller.

Example 6

Ali: median 7, range 3

Omar: median 5, range 10

Ali has the higher median,

so he scored more on average.

Ali has the smaller range,

so he is more consistent.

Tip: compare both, and say what each means.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint
Mean = total ÷ number of values
Smaller range = more consistent

1 Questions completed by 6 students: 12, 8, 15, 10, 9, 18
Work out the mean.

2 Children in 5 houses: 4, 2, 6, 1, 2
Work out the mean.

3 A 6th house in Q2 has 0 children.
Is the mean greater or less?

4 Find the median and range: 5, 2, 9, 4, 6, 8

5 Another set: median 4, range 14.
Compare it with Q4.

★ Which average best describes 3, 4, 4, 5, 24? Give a reason.

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1 Mean = 72 ÷ 6
= 12 questions

2 Mean = 15 ÷ 5
= 3

3 Less:
Mean = 15 ÷ 6
= 2.5

4 Median = (5 + 6) ÷ 2
= 5.5
Range = 9 – 2
= 7

5 Q4: higher median,
smaller range (more consistent)

★ Median = 4: the 24
pulls the mean up to 8

Plenary — Exit Ticket

PLENARY · 5 MIN

Consider the data: 6, 8, 5, 9, 6, 4, 6, 12. Without using your notes, find:

BRONZE Find the mode and the range.

SILVER Find the median.

GOLD Find the mean. Which average would you use? Why?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE

Mode = 6

Range = 12 – 4

= 8

SILVER

Median = (6 + 6) ÷ 2

= 6

GOLD

Mean = 56 ÷ 8

= 7

Use the median (6): the 12 pulls the mean up.

Exam Practice — Mode and Range

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint
Mode = most common value
Range = highest − lowest

Question 1

(2 marks)

The list shows the number of pets some students own.

2 0 3 2 1 2 4

(a) Write down the mode. (1)

.....

(b) Work out the range. (1)

.....

Question 2

(2 marks)

The heights, in cm, of 6 plants are:

23 31 18 27 31 25

(a) Write down the mode. (1)

..... cm

(b) Work out the range. (1)

..... cm

Exam Practice — The Median

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Put the values in order first
Median = middle value

Question 3

(2 marks)

The list shows the number of books 7 students read last month.

5 2 8 3 6 1 4

Find the median.

..... books

Question 4

(2 marks)

The times, in minutes, 6 students take to walk to school are:

12 7 15 10 8 20

Find the median time.

..... minutes

Exam Practice — Mean and Comparing

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint
Mean = total ÷ number of values
Smaller range = more consistent

Question 5

(3 marks)

The number of children in 5 houses:

3 1 5 2 4

(a) Work out the mean. (2)

.....

(b) A 6th house has 0 children. Is the mean for 6 houses greater or less? Give a reason. (1)

Question 6

(2 marks)

Points scored in 6 games:

Sami: 4 9 6 2 8 6

Omar: median 9, range 3

Use the median and range to compare Sami and Omar.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) Mode = 2 B1

(b) Range = 4 − 0

= 4 B1

Tip: 0 is a value: it counts as the lowest.

Question 2

(a) Mode = 31 cm B1

(b) Range = 31 − 18

= 13 cm B1

Tip: range = highest − lowest.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

1, 2, 3, 4, 5, 6, 8 — M1

Median = 4 books — A1

Tip: put the values in order first.

Question 4

7, 8, 10, 12, 15, 20

Median = (10 + 12) ÷ 2 — M1

= 11 minutes — A1

Tip: halfway between the two middle values.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) Total = 3 + 1 + 5 + 2 + 4

= 15

Mean = 15 ÷ 5 M1

= 3 A1

(b) Less: same total, 6 values B1

Tip: 15 ÷ 6 = 2.5, which is less than 3.

Question 6

Sami: median 6, range 7 B1

Omar: higher median, so B1

scored more on average

Omar: smaller range, so

more consistent

Tip: one comparison of each for both marks.